

Guidelines

Orthostatic Hypotension Guideline

Reference #: 004

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff

1. Table of Contents

Scope.....	1
Introduction	1
Risk Statement	2
Definitions	2
2.0 Causes/Contributing Factors of OH include ¹⁻¹³	2
3.0 Procedure ^{2,6,12,13}	3
4.0 Treatment	4
5.0 Discharge	4
6.0 Staff information.....	4
7.0 Compliance, monitoring and evaluation.....	4
8.0 Related Documents	5
9.0 References	5
Authorisation & Version Control.....	5

Introduction

Orthostatic Hypotension (OH) is significantly positively associated with falls in older adults, underpinning the clinical relevance to test for an orthostatic blood pressure drop and highlighting the need to investigate orthostatic hypotension treatment to potentially reduce falls^{1,2,3,4}. Among hospitalised patients about 60% of adults experience OH^{5,6} so it becomes important to identify, manage and treat.

Orthostatic intolerance symptoms are caused by transient cerebral hypoperfusion and typically occur upon standing during the designated time-period for the drop in blood pressure and resolve upon returning to baseline blood pressure or reclining⁴.

Management of orthostatic hypotension is aimed at improving quality of life and reducing symptoms rather than at normalizing blood pressure. Nonpharmacologic measures are the key to success.⁵

This guideline outlines, using evidenced based practice, the procedures and responsibilities of individuals and the health service organisation in relation to orthostatic hypotension. It aims to assist staff in the assessment, identification, and treatment of OH along with patient education.

Nursing, Physiotherapists, Occupational Therapists and Medical Officers should identify and investigate patients at risk of having and those that have OH.

Initial identification is via falls screening, falls history, clinical assessment, medication review and questioning of the patient/family/carer for any description of symptoms.

A lying and standing blood pressure is to be taken for the following patients:

- All patients admitted with a fall (where physically able to do so)
- All patients over 65 years of age
- All patients with delirium
- Patient complains of/exhibits any of the below symptoms; and/or
- Has been identified during a falls risk screening.
- The patient has a medical condition or takes medications that may potentially contribute to OH.

Refer to [SCGOPHCG Falls Website](#) for information and resources and education

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Other:	☐

Definitions

Orthostatic Hypotension	A drop in systolic blood pressure (BP) of 20mmHg or more (with or without symptoms)
	A drop to below 90mmHg on standing even if the drop is less than 20mmHg (with or without symptoms)
	A drop in diastolic BP of 10mmHg with symptoms (although clinically much less significant than a drop in systolic BP).

1.0 Symptoms of OH may include: ¹⁻¹³

- Falls
- Dizzy/lightheaded, especially when moving from lying or sitting to standing.
- Weakness
- Feeling faint or fainting
- Dimming/disturbance of vision
- Sweating and/or nausea
- Gait imbalance.
- Mild, momentary confusion
- Unexplained syncope
- Legs buckle.

2.0 Causes/Contributing Factors of OH include:¹⁻¹³

- Frailty and physical deconditioning
- Older adult age
- Standing or sitting up suddenly/quickly
- Hypertension
- Various medical conditions e.g., diabetes, anaemia
- Cardiac conditions including postural tachycardia syndrome, aortic stenosis, arrhythmias.
- Medications, e.g., diuretics, antidepressants, antihypertensives, antipsychotics, anticholinergics, antiparkinsonian medications, medicines for prostate hypertrophy
- Neurological conditions e.g., Parkinson's disease and some types of dementia
- Endocrinological conditions e.g., Addison's
- Spinal cord injuries

- Autonomic neuropathy
- Dehydration
- Reflex syncope e.g., vasovagal, neurocardiogenic, syncope associated with micturition and defecation.
- Exercise
- In the morning when the blood pressure is naturally lower
- Prolonged bed rest or inactivity
- Post prandial.

3.0 Procedure^{1,2,6,12,13,11}

The patients' falls risk screen, risk identification and interventions required must be reviewed and updated.

Taking a Lying & Standing Blood Pressure:

- Take twice a day for two days.
- Discuss with medical team and discontinue if no evidence/symptoms of OH.
- Continue taking lying and standing BP if there is evidence/symptoms.
- The blood pressure only needs to be taken at 1 minute standing unless symptoms persist in which case the standing blood pressure must be taken at 3 and 5 minutes.

Procedure:

1. Identify if you are going to need assistance to stand the patient and simultaneously measure the blood pressure (BP).
2. It is preferable to use a manual sphygmomanometer.

3. Explain procedure to the patient.

 After at least five minutes	 4. The first BP should be taken after the patient has been lying down for at least five minutes.
In the first minute	 5. The second BP should be taken within the first minute of the patient standing.

6. Document the result.

7. Symptoms of dizziness, light-headedness, vagueness, pallor, visual disturbance, feelings of weakness and palpitations should be documented on the patient's medical record.
8. Advise the patient of the results and, if the result is positive:
 - a. Inform the medical and nursing team.
 - b. Take immediate actions to prevent falls and/or unsteadiness.

Results:

A positive result is defined by:

- a. A drop in systolic BP of 20mmHg or more (with or without symptoms).
- b. A drop to below 90mmHg on standing even if the drop is less than 20mmHg (with or without symptoms).
- c. A drop in diastolic BP of 10mmHg with symptoms (although clinically much less significant than a drop in systolic BP).

If results are positive, repeat the procedure regularly or as per medical officer directives.

4.0 Treatment

Non-Pharmacological Treatment ^{2,5,6,8,10,12,113,11,}

- Review and update the MFRA (complete risk assessment and interventions)
- Provide education ([leaflet available](#)) to the patient and or carer
- Refer patient to [SCGOPHCG Falls CNC](#) for education and care planning.
- Commence [OH care advice](#)
- Patients should first sit when going from a supine to a standing position.
- Isometric exercises prior to going from supine to standing:
 - Lift alternate legs up and down, then move feet up and down.
 - Clench and unclench hands.
 - Buttock squeezing.
- Physical counter-manoevres when upright, increases venous return to the heart and enhances orthostatic tolerance:
 - Marching on the spot
 - Leg-crossing.
 - Standing on tiptoes
 - Muscle tensing
- Review of medications contributing to orthostatic hypotension and deprescribe offending medicines where possible
- Eating frequent, small meals (post prandial hypotension can occur)
- Full-length elastic stockings and consider use of [abdominal binders](#)
- Tilting the head of the bed up during the night – approx. 10 - 20°
- Salt intake may need increasing (on medical advice only)
- 2 litres of water a day (on medical advice and not for patients on fluid restrictions)
- Avoiding situations that trigger symptoms, such as standing for long periods.
- Lifestyle modification- minimal alcohol

Pharmacological Treatment ^{2,56,,7,, 10,12, 113:} Medical staff if possible, to stop medications potentially contributing to OH.

If pharmacological treatment is required, consider:

- First line: Fludrocortisone (watch for fluid retention, supine hypertension, and electrolyte disturbance)
- In resistant cases, seek specialist advice. Midodrine can be initiated by specialists only. Refer to [WA Statewide Medicines Formulary](#) for more information. (Timing of dosing is important - the last daily dose should be taken at least 4 hours before bedtime to prevent supine hypertension)

Investigations for medical staff to consider: When investigating OH, please consider:

- Urea and Electrolyte testing (query dehydration)
- Spot morning cortisol to rule out Addison's disease

5.0 Discharge

- Consider referral to Falls Clinic on discharge if patient has OH
- Ensure provided with [information leaflet](#) and education on how to manage OH

6.0 Staff information

See [Orthostatic Hypotension Falls Webpage](#) for education resources for all staff

7.0 Compliance, monitoring and evaluation

- Falls CNC annual audit.
- Monitor via SAC One incidents.

8.0 Related Documents

- [NMHS Falls Management Multidisciplinary Policy](#)
- [Falls Management Guideline HG018](#)
- [SCGOPHCG Clinical Handover Communicating for Safety Policy](#)
- [NMHS Clinical Incident Management Policy](#)
- [SCGOPHCG Cognitive Impairment Management Clinical Practice Guideline](#)
- [WA Language Policy](#)
- [WA Post Fall Multidisciplinary Management Guidelines for Western Australian Health Care Settings 2023](#)
- [National Safety and Quality Health Care Standards](#)

9.0 References

1. Mol A, Bui Hoang PTS, Sharmin S, Reijnierse EM, van Wezel RJA, Meskers CGM, Maier AB. Orthostatic Hypotension and Falls in Older Adults: A Systematic Review and Meta-analysis. J Am Med Dir Assoc. 2019 May;20(5):589-597.e5. doi: 10.1016/j.jamda.2018.11.003. Epub 2018 Dec 21. PMID: 30583909.
2. Gilani A, Juraschek SP, Belanger MJ, Vowles JE, Wannamethee SG. Postural hypotension. BMJ. 2021;373:n922. Published 2021 Apr 23. doi:10.1136/bmj.n922 Ali A, Ali NS, Waqas N, Bhan C, Iftikhar W, Sapna F, Jitidhar F, Cheema AM, Ahmad MQ, Nasir U, Sami SA, Zulfiqar A, Ahmed A. Management of Orthostatic Hypotension: A Literature Review. Cureus. 2018 Aug 20;10(8):e3166. doi: 10.7759/cureus.3166. PMID: 30357001; PMCID: PMC6197501.
3. Montero-Odasso M, et al. Task Force on Global Guidelines for Falls in Older Adults. World guidelines for falls prevention and management for older adults: a global initiative. Age Ageing. 2022 Sep 2;51(9):afac205. doi: 10.1093/ageing/afac205. PMID: 36178003; PMCID: PMC9523684
4. Fedorowski A, Ricci F, Hamrefors V, Sandau KE, Hwan Chung T, Muldowney JAS, Gopinathannair R, Olshansky B. Orthostatic Hypotension: Management of a Complex, But Common, Medical Problem. Circ Arrhythm Electrophysiol. 2022 Mar;15(3):e010573. doi: 10.1161/CIRCEP.121.010573. Epub 2022 Feb 25. Erratum in: Circ Arrhythm Electrophysiol. 2023 Aug;16(8):478. doi: 10.1161/HAE.0000000000000088. PMID: 35212554; PMCID: PMC9049902.
5. Christopoulos EM, Tran J, Hillebrand SL, Lange PW, Iseli RK, Meskers CGM, Maier AB. Initial orthostatic hypotension and orthostatic intolerance symptom prevalence in older adults: A systematic review. Int J Cardiol Hypertens. 2020 Dec 8;8:100071. doi: 10.1016/j.ijchy.2020.100071. PMID: 33884364; PMCID: PMC7803043.
6. Dani M, Dirksen A, Taraborrelli P, Panagopolous D, Torocastro M, Sutton R, Lim PB. Orthostatic hypotension in older people: considerations, diagnosis and management. Clin Med (Lond). 2021 May;21(3):e275-e282. doi: 10.7861/clinmed.2020-1044. PMID: 34001585; PMCID: PMC8140709.
7. Palma JA, Kaufmann H. Management of Orthostatic Hypotension. Continuum (Minneapolis Minn). 2020 Feb;26(1):154-177. doi: 10.1212/CON.0000000000000816. PMID: 31996627; PMCID: PMC7339914.
8. Subbarayan S, Myint PK, Martin KR, Abbraha I, Devkota S, O'Mahony D, Cruz-Jentoft AJ, Cherubini A, Soiza RL. Nonpharmacologic Management of Orthostatic Hypotension in Older People: A Systematic Review. The SENATOR ONTOP Series. J Am Med Dir Assoc. 2019 Sep;20(9):1065-1073.e3. doi: 10.1016/j.jamda.2019.03.032. Epub 2019 May 17. PMID: 31109911.
9. Kim MJ, Farrell J. Orthostatic Hypotension: A Practical Approach. Am Fam Physician. 2022 Jan 1;105(1):39-49. Erratum in: Am Fam Physician. 2022 Oct;106(4):365. PMID: 35029940.
10. Logan A, Marsden J, Freeman J, Kent B. Effectiveness of non-pharmacological interventions in treating orthostatic hypotension in the elderly and people with a neurological condition: a systematic review protocol. JBI Database System Rev Implement Rep 2017; 15(4):948-960
11. Tran J, Hillebrand SL, Meskers CGM, Iseli RK, Maier AB. Prevalence of initial orthostatic hypotension in older adults: a systematic review and meta-analysis. Age Ageing. 2021 Sep 11;50(5):1520-1528. doi: 10.1093/ageing/afab090. PMID: 34260686; PMCID: PMC8437071.
12. Juraschek SP, Daya N, Rawlings AM, Appel LJ, Miller ER, Windham G, Griswold ME, Heiss G, Selvin E. Comparison of Early versus Late Orthostatic Hypotension Assessment Times in Middle-Age Adults JAMA Intern Med. 2017 September 01; 177(9): 1316-1323. doi:10.1001/jamainternmed.2017.2937
13. Chislom P, Mahesan A. Orthostatic hypotension: pathophysiology, assessment, treatment and the paradox of supine hypertension. Internal Medicing Journal. 2016. London: Royal Australasian College of Physicians

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